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Third Semester B.E./B.Tech. Degree Examination, June/July 2025

Mathematics for Computer Science

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks, L: Bloom's level, C: Course outcomes.

Time: 3 hrs.		Note: 1. Answer any FIVE full questions.		2. M: Marks, L: Bloom's level, C: Course outcomes.	
Module – 1					
a.	The probability density function of a variate X is given by the following table:		M	L	C
		06	L2	CO1	

- c. The joint probability distribution for two random variables X and Y is given below :

Y \ X	-2	-1	4	5
1	0.1	0.2	0	0.3
2	0.2	0.1	0.1	0

Determine :

- The marginal distribution of X and Y.
- $E[X]$, $E[Y]$, $E[XY]$
- $COV(X, Y)$

07

L2

CO3

OR

- Q.4 a. Find the fixed probability vector of the regular stochastic matrix

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

06

L2

CO3

- b. A habitual gambler is a member of two clubs A and B. He visits either of the clubs every day for playing cards. He never visits club A on two consecutive days. But if he visits club B on a particular day then the next day he is as likely to visit club B or club A. If the person had visited club B on Monday, find the probability that he visits club A on Thursday.

07

L3

CO3

- c. The joint probability distribution of two discrete random variable X and Y is given by, $f(x, y) = K(2x+y)$ where x, y are integers such that $0 \leq x \leq 2$, $0 \leq y \leq 2$. Find
- K
 - Marginal probability distribution of X and Y
 - $P(X \geq 1, Y \leq 2)$

07

L2

CO2

Module – 3

- Q.5 a. Define : (i) Null hypothesis (ii) Significance level
(iii) Type I and Type II errors

06

L1

CO4

- b. A die was thrown 1200 times and the number 6 was obtained 236 times. Can the die be considered fair at 1% level of significance?

07

L3

CO4

- c. One type of aircraft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significance difference in the two types of aircrafts so far as engine defects are concerned?

07

L3

CO4

OR

- Q.6 a. Explain (i) Sampling distribution (ii) Statistical hypothesis
(iii) Testing of hypothesis

06

L1

CO4

- b. A coin is tossed 1000 times and head turns up 540 times. Decide on the hypothesis that the coin is unbiased at 1% level of significance.

07

L3

CO4

- c. In an exit poll enquiry it was revealed that 600 voters in one locality and 400 voters from another locality favoured 55% and 48% respectively a particular party to come to power. Test the hypothesis that there is a difference in the locality in respect of the opinion at 5% level of significance.

07

L3

CO4



Module - 4

Q.7	a.	(i) State central limit theorem. (ii) What is 95% and 99% confidence limits for unknown mean μ of a random sample of size n ?	06	L1	CO5
	b.	Certain tubes manufactured by a company have mean life of 800 hours and standard deviation of 60 hours. Using central limit theorem, find the probability that a random sample of 16 tubes taken from the group will have mean life time. (i) Between 790 hours and 810 hours. (ii) Less than 785 hours (iii) More than 820 hours Given $A(0.67) = 0.2486$, $A(1) = 0.3413$, $A(1.33) = 0.4082$	07	L2	CO4
	c.	Two Samples of sizes 9 and 8 give the sum of squares of deviations from their respective means equal to 160 inches ² and 91 inches ² respectively. Can these be regarded as drawn from the same normal population? Use 5% points of significance for F.	07	L2	CO4

OR

Q.8	a.	A random sample of 400 items chosen from an infinite population is found to have a mean of 82 and a standard deviation of 18. Find the 95% confidence limits for the mean of the population from which the sample is drawn.	06	L1	CO5														
	b.	A mechanist is making engine parts with axle diameter of 0.7 inch. A random sample of 10 parts shows mean diameter 0.742 inch with standard deviation of 0.04 inch. On the basis of this sample, would you say that the work is interior, given $t_{0.05} = 2.262$ for degrees of freedom = 9.	07	L2	CO4														
	c.	A set of five similar coins is tossed 320 times and result is,<table><tr><td>Number of heads</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Frequency</td><td>6</td><td>27</td><td>72</td><td>112</td><td>71</td><td>32</td></tr></table> Test the hypothesis that the data follows Binomial distribution, (Given : $\chi^2_{0.05} = 11.07$ for d.f = 5)	Number of heads	0	1	2	3	4	5	Frequency	6	27	72	112	71	32	07	L3	CO4
Number of heads	0	1	2	3	4	5													
Frequency	6	27	72	112	71	32													

Module - 5

Module – 5																																		
a.	Set up ANOVA table for the following information relating to three drugs testing to judge the effectiveness in reducing blood pressure for three different groups of people.			10	L3	CO6																												
<table><tr><th rowspan="2">Group of people</th><th colspan="3">Drug</th></tr><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td rowspan="2">A</td><td>14</td><td>10</td><td>11</td></tr><tr><td>15</td><td>9</td><td>11</td></tr><tr><td rowspan="2">B</td><td>12</td><td>7</td><td>10</td></tr><tr><td>11</td><td>8</td><td>11</td></tr><tr><td rowspan="2">C</td><td>10</td><td>11</td><td>8</td></tr><tr><td>11</td><td>11</td><td>7</td></tr></table>				Group of people	Drug			X	Y	Z	A	14	10	11	15	9	11	B	12	7	10	11	8	11	C	10	11	8	11	11	7			
Group of people	Drug																																	
	X	Y	Z																															
A	14	10	11																															
	15	9	11																															
B	12	7	10																															
	11	8	11																															
C	10	11	8																															
	11	11	7																															
Do the drugs act differently? Are the different groups of people affected differently? Is the interaction term significant? Answer the above questions taking a significant level of 5%. Given $F(2, 9) = 4.26$, $F(4, 9) = 3.63$																																		

- b. Present your conclusions after doing analysis of variance to the following results of the Latin square design experiment conducted in respect of five fertilizers which were used on plots of different fertility. Given

A	B	C	D	E
16	10	11	9	9
E	C	A	B	D
10	9	14	12	11
B	D	E	C	A
15	8	8	10	18
D	E	B	A	C
12	6	13	13	12
C	A	D	E	B
13	11	10	7	14

OR

- Q.10 a. It is desired to compare three hospitals with regards to the number of deaths per month. A sample of death records were selected from the records of each hospital and the number of deaths was as given below. From these data, use ANOVA and suggest a difference in the number of deaths per month among the three hospitals. Given at 5% level, $F_{2,12} = 3.89$.

Hospitals		
A	B	C
3	6	7
4	3	3
3	3	4
5	4	6
0	4	5

- b. Analyze and interpret the following statistics concerning output of wheat per field obtained as a result of experiment conducted to test four varieties of wheat viz. A, B, C and D under a Latin square design.

C	B	A	D
25	23	20	20
A	D	C	B
19	19	21	18
B	A	D	C
19	14	17	20
D	C	B	A
17	20	21	15

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Mathematics – III for Computer Science Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. VTU Mathematics Hand Book is permitted.
 3. M : Marks, L: Bloom's level, C: Course outcomes.

marks, L: Bloom's level, C: Course outcomes.

Module – 1

Q.1	a.	A random variable x has the following prob. density function for various values of x .	M	L	C																		
		<table> <tr> <td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr> <tr> <td>$P(x)$</td><td>0</td><td>k</td><td>$2k$</td><td>$2k$</td><td>$3k$</td><td>k^2</td><td>$2k^2$</td><td>$7k^2+k$</td></tr> </table> Find the value of k and evaluate $P(x < 6)$, $P(3 < x \leq 6)$ and $(x \geq 6)$.	x	0	1	2	3	4	5	6	7	$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$	07	L2	CO1
x	0	1	2	3	4	5	6	7															
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$															
	b.	Derive the mean and variance of Poisson distribution.	06	L2	CO2																		
	c.	In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for? (i) less than 10 minutes (ii) more than 10 minutes and (iii) between 10 and 12 minutes.	07	L3	CO2																		

OR

Q.2	a.	The probability density function of	M	L	C
		$f(x) = \begin{cases} Kx^2, & -3 < x < 3 \\ 0, & \text{elsewhere} \end{cases}$ Find the value of K and evaluate (i) $P(x < 2)$, $P(x > 1)$ (ii) $P(1 \leq x \leq 2)$	07	L3	CO1
	b.	When a coin is tossed 4 times, find the probability of getting (i) exactly one head (ii) at least three heads and (iii) less than two heads.	06	L2	CO2
	c.	The marks of 1000 students in an examination follows a normal distribution with mean > 0 and S.D 5. Find the number of students whose marks will be (i) less than 65 (ii) more than 75 and (iii) between 65 and 75.	07	L2	CO2

Module – 2

Q.3	a.	If the joint probability distribution of x and y is given by	M	L	C
		$f(x, y) = \frac{1}{30}(x + y), \text{ for } x = 0, 1, 2, 3; y = 0, 1, 2$ Find (i) $P(x \leq 2, y = 1)$ (ii) $P(x > y)$	07	L2	CO2
	b.	Find the unique fixed probability vector of	06	L2	CO3
		$P = \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix}$			
	c.	Three boys A, B and C are throwing a ball to each other. A always throw the ball to B. B always throw the ball to A and C is just as likely to throw the ball to A as to B. Find the probability that C has the ball after three throws, if C starts the game.	07	L3	CO3

1 of 3

OR

Q.4	a.	The joint prob. distribution for the following data, find $E(x)$ and $E(y)$.	07	L2	CO2																				
		<table><tr><td>Y</td><td>-2</td><td>-1</td><td>4</td><td>5</td></tr><tr><td>X</td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>0.1</td><td>0.2</td><td>0.0</td><td>0.3</td></tr><tr><td>2</td><td>0.2</td><td>0.1</td><td>0.1</td><td>0</td></tr></table>	Y	-2	-1	4	5	X					1	0.1	0.2	0.0	0.3	2	0.2	0.1	0.1	0			
Y	-2	-1	4	5																					
X																									
1	0.1	0.2	0.0	0.3																					
2	0.2	0.1	0.1	0																					
	b.	Show that the matrix $P = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$ is a regular stochastic matrix.	06	L2	CO3																				
	c.	A gambler's luck follows pattern. If he wins a game the prob. of winning the next game is 0.6. However, if he loses a game, the prob. of losing the next game is 0.7. There is an even chance of the gambler winning the first game. What is the prob. of he winning the second game.	07	L3	CO3																				

Module – 3

Q.5	a.	Define (i) Null hypothesis (ii) A statistic (iii) Standard error (iv) Level of significance (v) Test of significance.	07	L1	CO4
	b.	A coin was tossed 400 times and head turned up 216 times. Test the hypothesis that the coin is unbiased at 5% LOS.	06	L3	CO4
	c.	In a city A 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant at 5% significance level?	07	L3	CO5

OR

Q.6	a.	Explain the following terms: (i) Type-I and Type-II errors (ii) Statistical hypothesis (iii) Critical region (iv) Alternate hypothesis	07	L1	CO4
	b.	The average marks in Engg. Maths of a sample of 100 students was 51 with S.D 6 marks. Could this have been a random sample from a population with average marks 50?	06	L2	CO5
	c.	One type of aircraft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significance difference in the two types of aircrafts so far as engine defects are concerned? Test at 0.05 significance level.	07	L3	CO4

Module – 4

Q.7	a.	State central limit theorem. Use the theorem to evaluate $P(50 < x < 56)$ where $\bar{x}$ represents the mean of a random sample of size 100 from an infinite population with mean $\mu = 53$ and variance $\sigma^2 = 400$.	07	L2	CO4												
	b.	Suppose that 10, 12, 16, 19 is a sample taken from a normal population with variance 6.25. Find 95% confidence interval for the population mean. Given that $Z(0.15) = 0.0596$.	06	L2	CO5												
	c.	Fit a Poisson distribution to the following data and test for goodness of fit at 5% LOS.	07	L3	CO5												
		<table border="1"> <tr> <td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr> <tr> <td>f</td><td>419</td><td>352</td><td>154</td><td>56</td><td>19</td></tr> </table>	x	0	1	2	3	4	f	419	352	154	56	19			
x	0	1	2	3	4												
f	419	352	154	56	19												

OR			
Q.8	a.	Height of a random sample of 50 college student showed a mean of 174.5 cms and a S.D 6.9 cms. Construct 99% confidence limits for the mean height of all college students.	07 L2 CO4
	b.	A random sample of 10 boys had the following IQ : 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. DO these data support the assumption of a population mean IQ of 100 (at 5% LOS)?	06 L3 CO5
	c.	The theory predicts the proportions of beans in the four groups, G_1, G_2, G_3, G_4 should be in the ratio 9 : 3 : 3 : 1. In experiment with 1600 beans the numbers in the groups were 882, 313, 287 and 118. Does the experimental support the theory.	07 L3 CO5

Module - 5

Module - 5

Q.9	a.	The varieties of wheat A, B, C were shown in four plots each and the following yields in quintals per acre were obtained. <table><tr><td>A</td><td>8</td><td>4</td><td>6</td><td>7</td></tr><tr><td>B</td><td>7</td><td>6</td><td>5</td><td>3</td></tr><tr><td>C</td><td>2</td><td>5</td><td>4</td><td>4</td></tr></table> Test the significance of difference between the yields of varieties, given that 5% tabulated value of $F = 4.26$ with (2, 9) d.f. Set up one-way ANOVA and using direct method.	A	8	4	6	7	B	7	6	5	3	C	2	5	4	4	10	L3	CO6										
A	8	4	6	7																										
B	7	6	5	3																										
C	2	5	4	4																										
	b.	Present your conclusion after doing ANOVA to the following results of the Latin-square design conducted in respect of five fertilizers which were used on plots of different fertility. <table><tr><td>A(16)</td><td>B(10)</td><td>C(11)</td><td>D(9)</td><td>E(9)</td></tr><tr><td>E(10)</td><td>C(9)</td><td>A(14)</td><td>B(12)</td><td>D(11)</td></tr><tr><td>B(15)</td><td>D(8)</td><td>E(8)</td><td>C(10)</td><td>A(18)</td></tr><tr><td>D(12)</td><td>E(6)</td><td>B(13)</td><td>A(13)</td><td>C(12)</td></tr><tr><td>C(13)</td><td>A(11)</td><td>D(10)</td><td>E(7)</td><td>B(14)</td></tr></table>	A(16)	B(10)	C(11)	D(9)	E(9)	E(10)	C(9)	A(14)	B(12)	D(11)	B(15)	D(8)	E(8)	C(10)	A(18)	D(12)	E(6)	B(13)	A(13)	C(12)	C(13)	A(11)	D(10)	E(7)	B(14)	10	L3	CO6
A(16)	B(10)	C(11)	D(9)	E(9)																										
E(10)	C(9)	A(14)	B(12)	D(11)																										
B(15)	D(8)	E(8)	C(10)	A(18)																										
D(12)	E(6)	B(13)	A(13)	C(12)																										
C(13)	A(11)	D(10)	E(7)	B(14)																										

OR

Q.10	a.	Set up two-way ANOVA table for the data given below, using coding method subtracting 40 from the given numbers. <table border="1"><tr><th rowspan="2">Pieces of land</th><th colspan="4">Treatment</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>P</td><td>45</td><td>40</td><td>38</td><td>37</td></tr><tr><td>Q</td><td>43</td><td>41</td><td>45</td><td>38</td></tr><tr><td>R</td><td>39</td><td>39</td><td>41</td><td>41</td></tr></table>	Pieces of land	Treatment				A	B	C	D	P	45	40	38	37	Q	43	41	45	38	R	39	39	41	41	10	L3	CO6
Pieces of land	Treatment																												
	A	B	C	D																									
P	45	40	38	37																									
Q	43	41	45	38																									
R	39	39	41	41																									
	b.	There are three main brands of a certain power. A set of its 120 sales is examined and found to be allocated among four groups (A, B, C, D) and brands (I, II, III) as follows: <table border="1"><tr><th rowspan="2">Brands</th><th colspan="4">Groups</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>I</td><td>0</td><td>4</td><td>8</td><td>15</td></tr><tr><td>II</td><td>5</td><td>8</td><td>13</td><td>6</td></tr><tr><td>III</td><td>18</td><td>19</td><td>11</td><td>13</td></tr></table> Is there any significant difference in brands preference? Answer at 5% level, using one-way ANOVA. Take 10 as the code value to subtract it from all given values.	Brands	Groups				A	B	C	D	I	0	4	8	15	II	5	8	13	6	III	18	19	11	13	10	L3	CO6
Brands	Groups																												
	A	B	C	D																									
I	0	4	8	15																									
II	5	8	13	6																									
III	18	19	11	13																									

Model Question Paper-I (CBCS Scheme)

USN

Third Semester B.E Degree Examination

MATHEMATICS FOR CS ENGINEERING STREAM (BCS301)

TIME: 03 Hours

Max.Marks:100

Note: (i) Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
(ii) Statistical tables and Mathematics Formula handbooks are allowed.

Module -1			M	L	C
Q.01	a	A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives. Find the mean and variance of this distribution.	6	L2	CO1
	b	In a factory producing blades, the probability of any blade being defective is 0.002. If blades are supplied in packets of 10, Using Poisson distribution determine the number of packets containing i. No defective ii. One defective and iii. Two defective blades respectively in a consignment of 10000 packets	7	L2	CO2
	c	If the mileage (in thousands of miles) of a certain radial tyre is a random variable with exponential distribution with mean 40,000 miles. Determine the probability that the tyre will last i. At least 20,000 miles ii. At most 30,000 miles	7	L3	CO2
OR					
Q.02	a	The density function of a random variable X is given by $f(x) = \begin{cases} k\sqrt{x}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$ i. Find k ii. Find the cdf F(x) and use it to evaluate $P[0.3 < X < 0.6]$	6	L2	CO1
	b	Find the mean and variance of Binomial distribution.	7	L2	CO1

	c	In a test on 2000 electric bulbs, it was found that the life of a particular make, was normally distributed with an average life 2040 hours and standard deviation of 60 hours. Estimate the number of bulbs likely to burn for - More than 2150 hours - Less than 1950 hours - In between 1920 and 2160 hours	7	L3	CO2
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Module-2

Q. 03	a	The joint probability distribution of two random variables X and Y is <table><tr><td> X </td><td>-4</td><td>2</td><td>7</td></tr><tr><td>Y</td><td>1</td><td>1/8</td><td>1/4</td><td>1/8</td></tr><tr><td></td><td>5</td><td>1/4</td><td>1/8</td><td>1/8</td></tr></table> Find the marginal distributions. Obtain the correlation coefficient between X and Y.	X	-4	2	7	Y	1	1/8	1/4	1/8		5	1/4	1/8	1/8	6	L2	CO1
X	-4	2	7																
Y	1	1/8	1/4	1/8															
	5	1/4	1/8	1/8															

	b	Find the unique fixed probability vector of $P = \begin{pmatrix} 0 & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{pmatrix}$	7	L2	CO3
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	c	Every year, a man trades his car for a new car. If he has a Maruti, he trades it for an Ambassador. If he has an Ambassador, he trades it for Santro. However, if he had a Santro, he is just as likely to trade it for a new Santro as to trade it for a Maruti or an Ambassador. In 2000 he bought his first car which was a Santro. Find the probability that he has 2002 Santro 2002 Maruti 2003 Ambassador 2003 Santro	7	L3	CO3
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OR

Q.04	a	The joint probability distribution of two random variables X and Y is <table><tr><td> X </td><td>-3</td><td>2</td><td>4</td></tr><tr><td>Y</td><td></td><td></td><td></td></tr><tr><td>1</td><td>0.1</td><td>0.2</td><td>0.2</td></tr><tr><td>3</td><td>0.3</td><td>0.1</td><td>0.1</td></tr></table>	X	-3	2	4	Y				1	0.1	0.2	0.2	3	0.3	0.1	0.1	6	L2	CO1
X	-3	2	4																		
Y																					
1	0.1	0.2	0.2																		
3	0.3	0.1	0.1																		
		i. Are X and Y independent?																			

		ii. Evaluate $P[Y \leq 2]$ iii. Evaluate $P[X + Y \leq 2]$										
	b	Define Probability Vectors, Stochastic matrices, Regular stochastic matrix, stationary distribution and absorbing state of Markov chain.	7	L1	CO3							
	c	A salesman's territory consists of 3 cities A, B and C. He never sells in the same city on successive days. If he sells in city A, then the next day he sells in city B. However, if he sells in either B or C, then the next day he is twice as likely to sell in city A as in other city. In long run, how often does he sell in each of the cities.	7	L3	CO3							
Module-3												
Q. 05	a	Define Null hypothesis, significance level, critical region, Type-I and type-II errors in a statistical test.	6	L1	CO4							
	b	A coin was tossed 400 times and head turned up 216 times. Test the hypothesis that the coin is unbiased at 5% level of significance.	7	L3	CO4							
	c	In a city A 20 % of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant at 5% significance level	7	L3	CO4							
OR												
Q. 06	a	Explain the following terms i. Standard error ii. Statistical hypothesis iii. Critical region of a statistical test iv. Test of significance	6	L1	CO4							
	b	A die was thrown 9000 times and a throw of 5 or 6 was obtained 3240 times. On the assumption of random throwing, do the data indicate an unbiased die at 1% level of significance.?	7	L3	CO4							
	c	In a sample of 600 men from a certain city, 450 are found smokers. In another sample of 900 men from another city, 450 are smokers. Do the indicate that the cities are significantly different with respect to the habit of smoking among men. Test at 5% significance level.	7	L3	CO4							
Module-4												
Q. 07	a	State Central limit theorem. Use the theorem to evaluate $P[50 < \bar{X} < 56]$ where $\bar{X}$ represents the mean of a random sample of size 100 from an infinite population with mean $\mu = 53$ and variance $\sigma^2 = 400$	6	L2	CO5							
	b	A random sample of size 25 from a normal distribution $N(\mu, \sigma^2 = 4)$ yields, sample mean $\bar{X} = 78.3$. Obtain a 99% confidence interval for μ .	7	L2	CO5							
	c	A survey of 320 families with 5 children each revealed the following distribution. <table><tr><td>No. of boys</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr></table>	No. of boys	5	4	3	2	1	0	7	L3	CO4
No. of boys	5	4	3	2	1	0						

		<table><tr><td>No. of girls</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>No. of families</td><td>14</td><td>56</td><td>110</td><td>88</td><td>40</td><td>12</td></tr></table> Is the result consistent with the hypothesis that male and female births are equally probable at 5% level of significance.	No. of girls	0	1	2	3	4	5	No. of families	14	56	110	88	40	12																					
No. of girls	0	1	2	3	4	5																															
No. of families	14	56	110	88	40	12																															
OR																																					
Q. 08	a	A random sample of size 64 is taken from an infinite population having mean 112 and variance 144. Using central limit theorem, find the probability of getting the sample mean $\bar{X}$ greater than 114.5	6	L2	CO5																																
	b	Let the observed value of the mean $\bar{X}$ of a random sample of size 20 from a normal distribution with mean μ and variance $\sigma^2 = 80$ be 81.2. Find a 90% and a 95% confidence intervals for μ .	7	L2	CO5																																
	c	The nine items of a sample have the following values: 45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of these differ significantly from the assumed mean of 47.5 at 5% significance level?	7	L3	CO4																																
Module-5																																					
Q. 09	a	Three different kinds of food are tested on three groups of rats for 5 weeks. The objective is to check the difference in mean weight (in grams) of the rats per week. Apply one-way ANOVA using a 0.05 significance level to the following data: <table><tr><td>Food 1</td><td>8</td><td>12</td><td>19</td><td>8</td><td>6</td><td>11</td></tr><tr><td>Food 2</td><td>4</td><td>5</td><td>4</td><td>6</td><td>9</td><td>7</td></tr><tr><td>Food 3</td><td>11</td><td>8</td><td>7</td><td>13</td><td>7</td><td>9</td></tr></table>	Food 1	8	12	19	8	6	11	Food 2	4	5	4	6	9	7	Food 3	11	8	7	13	7	9	10	L3	CO6											
Food 1	8	12	19	8	6	11																															
Food 2	4	5	4	6	9	7																															
Food 3	11	8	7	13	7	9																															
	b	Analyze and interpret the following statistics concerning output of wheat per field obtained as a result of experiment conducted to test four varieties of wheat viz. A, B, C and D under a Latin- square design <table><tr><td>C</td><td>B</td><td>A</td><td>D</td></tr><tr><td>25</td><td>23</td><td>20</td><td>20</td></tr><tr><td>A</td><td>D</td><td>C</td><td>B</td></tr><tr><td>19</td><td>19</td><td>21</td><td>18</td></tr><tr><td>B</td><td>A</td><td>D</td><td>C</td></tr><tr><td>19</td><td>14</td><td>17</td><td>20</td></tr><tr><td>D</td><td>C</td><td>B</td><td>A</td></tr><tr><td>17</td><td>20</td><td>21</td><td>15</td></tr></table>	C	B	A	D	25	23	20	20	A	D	C	B	19	19	21	18	B	A	D	C	19	14	17	20	D	C	B	A	17	20	21	15	10	L3	CO6
C	B	A	D																																		
25	23	20	20																																		
A	D	C	B																																		
19	19	21	18																																		
B	A	D	C																																		
19	14	17	20																																		
D	C	B	A																																		
17	20	21	15																																		

OR

Q. 10 a

Set up an analysis of variance table for the following per acre production data for three varieties of wheat, each grown on 4 plots and state if the variety differences are significant at 5% significant level

Per acre production data			
Plot of land	Variety of Wheat		
	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

10

L3

CO6

b

Set up ANOVA table for the following information relating to three drugs testing to judge the effectiveness in reducing blood pressure for three different groups of people

Group of people	Drug		
	X	Y	Z
A	14	10	11
	15	9	11
B	12	7	10
	11	8	11
C	10	11	8
	11	11	7

10

L3

CO6

Do the drugs act differently?

Are the different groups of people affected differently?

Is the interaction term significant?

Answer the above questions taking a significant level of 5%

Model Question Paper-I with effect from 2023-24 (CBCS Scheme)

USN

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Third Semester B.E. Degree Examination
Mathematics for Computer Science

TIME: 03 Hours

Max. Marks: 100

- Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
 02. Statistical tables and Mathematics formulae handbooks are allowed

Module -1			Bloom's Taxonomy Level	Marks																
Q.01	a	The probability distribution function of variate X is given by the following table; <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>P(x)</td><td>k</td><td>3k</td><td>5k</td><td>7k</td><td>9k</td><td>11k</td><td>13k</td></tr></table> i) Find the value of k, ii) $P(x \geq 5)$ & iii) $P(3 < x \leq 6)$.	x	0	1	2	3	4	5	6	P(x)	k	3k	5k	7k	9k	11k	13k	L2	6
x	0	1	2	3	4	5	6													
P(x)	k	3k	5k	7k	9k	11k	13k													
	b	If the probability of a bad reaction from a certain injection is 0.001, determine the chance that more than two of 2000 individuals will have a bad reaction.	L3	7																
	c	Find the mean and standard deviation of Poisson's distribution.	L2	7																
OR																				
Q.02	a	The probability of a pen manufactured by a factory be defective is $1/10$. If 12 such pens are manufactured, what is the probability that i) exactly 2 are defective, ii) at least 2 are defective, iii) none of them are defective.	L3	6																
	b	Determine the value of k, so that the function $f(x) = k(x^2 + 4)$ for $x = 0, 1, 2, 3$ can serve as a probability distribution of the discrete random variable X: Also find i) $P(0 < x \leq 2)$ and ii) $P(x \geq 1)$.	L2	7																
	c	Find the mean and standard deviation of Binomial distribution.	L2	7																
Module-2																				
Q.03	a	The joint probability distribution of discrete random variables X & Y are as follows; <table><tr><td>X\Y</td><td>-3</td><td>2</td><td>4</td></tr><tr><td>1</td><td>0.1</td><td>0.2</td><td>0.2</td></tr><tr><td>2</td><td>0.3</td><td>0.1</td><td>0.1</td></tr></table> Then i) determine marginal distribution of X & Y, ii) show that X & Y are dependent.	X\Y	-3	2	4	1	0.1	0.2	0.2	2	0.3	0.1	0.1	L2	6				
X\Y	-3	2	4																	
1	0.1	0.2	0.2																	
2	0.3	0.1	0.1																	
	b	Determine the value of k so that the function $f(x, y) = k x - y $, for $x = -2, 0, 2$; $y = -2, 3$ represents joint probability distribution of the random variables X and Y. Also determine $\text{cov}(X, Y)$.	L2	7																
	c	Three boys X, Y, Z are throwing a ball to each other. X always throws the ball to Y & Y always throws the ball to Z. But Z is just as likely to throw the ball to Y or as to X. Write TPM if Z is the first person to throw the ball, find the probability that X has the ball after fourth throw.	L3	7																
OR																				

Q.04	a	Given the following joint distribution of the random variable X & Y, <table><tr><td>X\Y</td><td>-2</td><td>-1</td><td>4</td><td>5</td></tr><tr><td>1</td><td>0.1</td><td>0.2</td><td>0</td><td>0.3</td></tr><tr><td>2</td><td>0.2</td><td>0.1</td><td>0.1</td><td>0</td></tr></table> Determine the marginal probability distributions of X & Y. Also compute i) Expectations of X, Y & XY, ii) Covariance of X & Y, iii) Correlation of X & Y.	X\Y	-2	-1	4	5	1	0.1	0.2	0	0.3	2	0.2	0.1	0.1	0	L2	6
	X\Y	-2	-1	4	5														
1	0.1	0.2	0	0.3															
2	0.2	0.1	0.1	0															
b	The joint probability distribution of random variables X & Y are as follows. <table><tr><td>x\y</td><td>-4</td><td>2</td><td>7</td></tr><tr><td>1</td><td>1/8</td><td>1/4</td><td>1/8</td></tr><tr><td>5</td><td>1/4</td><td>1/8</td><td>1/8</td></tr></table> then determine i) marginal distribution of X & Y, ii) E(X), E(Y) & E(XY), iii) COV(X,Y), iv) $\rho(X,Y)$.	x\y	-4	2	7	1	1/8	1/4	1/8	5	1/4	1/8	1/8	L2	7				
x\y	-4	2	7																
1	1/8	1/4	1/8																
5	1/4	1/8	1/8																
c	The students study habits are as follows; If he studies on one night, he is 60% sure not to study on next night. On the other hand, if he does not study on to night, he is 80% sure to study next night. Write the transition probability matrix for his chain of study. In the long run how often does he study? Suppose he studies on Monday night, what is the probability that he does not study on Friday night?	L3	7																
Module-3																			
Q.05	a	A die was thrown 9000 times and throw of 5 or 6 was obtained 3240 times on the assumption of random throwing do the data indicate an unbiased die?	L3	6															
	b	Before an increase in excise duty on tea 400 people out of a sample 500 persons were found to be tea drinkers. After an increase in duty 400 people were tea drinkers in a sample of 600 people. Using standard error of proportion, state whether there is a significant decrease in the consumption of tea for 95% and 99% level of significance?	L3	7															
	c	A survey was conducted in a slum locality of 2000 families by selecting a sample of size 800 families. It was revealed that 180 families were illiterate. Find the probable limits of the illiterate families in the population of 2000.	L3	7															
OR																			
Q.06	a	In 324 throws a die an odd number turned up 181 times. Is it reasonable to think that the die is an unbiased one?	L3	6															
	b	The mean weight obtained from a random sample of size 100 is 64gms. The standard deviation of the weight distribution of the population is 3gms. Test the statement that the mean weight of the population is 67gms at 5% level of significance. Also set up 99% confidence limits of the mean weight of the population.	L3	7															
	c	In a sample of 100 people in this city, the average income was Rs. 210, with a standard deviation of Rs. 10. For another sample of 150 persons, the average income was Rs. 220, with a standard deviation of Rs. 12. The standard deviation of the incomes of the people of the city was Rs. 11. Test whether there is any significant difference between the average incomes of the localities.	L3	7															

Module-4

Q.07	a	An experiment on Pea breeding the following frequency of seeds were obtained <table><tr><td>Round & Yellow</td><td>Wrinkled & Yellow</td><td>Round & Green</td><td>Wrinkled & Green</td><td>Total</td></tr><tr><td>315</td><td>101</td><td>108</td><td>32</td><td>556</td></tr></table> Theory predicts that the frequencies should be in the proportions 9:3:3:1. Examine the correspondence between theory and experiment ($\chi^2_{0.05}=7.815$).	Round & Yellow	Wrinkled & Yellow	Round & Green	Wrinkled & Green	Total	315	101	108	32	556	L3	6														
Round & Yellow	Wrinkled & Yellow	Round & Green	Wrinkled & Green	Total																								
315	101	108	32	556																								
	b	Use the Central Limit theorem to evaluate $P[50 < \bar{X} < 56]$ where $\bar{X}$ represents the mean of a random sample of size 100 from an infinite population with mean $\mu = 53$ and variance $\sigma^2 = 400$ (Given, $A(1.5) = 0.4332$).	L2	7																								
	c	Ten individuals are chosen at random from a population and their heights in inches found to be 63, 63, 66, 67, 68, 69, 70, 70, 71 and 71. Test the hypothesis that the mean height of the universe is 66 inches. ($t_{0.05} = 2.262$ for 9 d.f.).	L3	7																								
OR																												
Q.08	a	The following table shows the runs scored by two batsmen can it be said that the performance of batsman A is more consistent than the performance of batsman B? Use 1% level of significance ($F_{0.01,4,7} = 7.85$). <table><tr><td>Batsman-A</td><td>40</td><td>50</td><td>35</td><td>25</td><td>60</td><td>70</td><td>65</td><td>55</td></tr><tr><td>Batsman-B</td><td>60</td><td>70</td><td>40</td><td>30</td><td>50</td><td></td><td></td><td></td></tr></table>	Batsman-A	40	50	35	25	60	70	65	55	Batsman-B	60	70	40	30	50				L3	6						
Batsman-A	40	50	35	25	60	70	65	55																				
Batsman-B	60	70	40	30	50																							
	b	The following table gives the number of aircraft accidents that occurred during the various days of the week. Find whether the accidents are uniformly distributed over the week? <table><tr><td>Days</td><td>Sun</td><td>Mon</td><td>Tue</td><td>Wed</td><td>Thu</td><td>Fri</td><td>Sat</td><td>Total</td></tr><tr><td>Accidents</td><td>14</td><td>16</td><td>8</td><td>12</td><td>11</td><td>9</td><td>14</td><td>84</td></tr></table> Given that $\chi^2_{0.05} = 12.59$.	Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total	Accidents	14	16	8	12	11	9	14	84	L3	7						
Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total																				
Accidents	14	16	8	12	11	9	14	84																				
	c	Consider the sample consisting of nine numbers 45, 47, 50, 52, 48, 47, 49, 53, 51. The sample is drawn from a population whose mean is 47.5. Find whether the sample mean differs significantly from the population mean at 5% level of significance ($t_{0.05}$ for 8 d.f. = 2.31).	L2	7																								
Module-5																												
Q.09	a	A manufacturing company has purchase three new machines of different brands and wishes to determine whether one of them is faster than the others in producing a certain output, 5 hourly production figures are obtained at random from each other machine and the results are given below; <table><tr><td>Observation</td><td>A</td><td>B</td><td>C</td></tr><tr><td>1</td><td>25</td><td>31</td><td>24</td></tr><tr><td>2</td><td>30</td><td>39</td><td>30</td></tr><tr><td>3</td><td>36</td><td>38</td><td>28</td></tr><tr><td>4</td><td>38</td><td>42</td><td>25</td></tr><tr><td>5</td><td>31</td><td>35</td><td>28</td></tr></table> Use ANOVA and determine whether the machines are significantly	Observation	A	B	C	1	25	31	24	2	30	39	30	3	36	38	28	4	38	42	25	5	31	35	28	L3	10
Observation	A	B	C																									
1	25	31	24																									
2	30	39	30																									
3	36	38	28																									
4	38	42	25																									
5	31	35	28																									

different in their mean speed. Given at 5% level $F_{2,12} = 3.89$.

- b Set up an analysis of variance table for the following per acre production data for three varieties of wheat, each grown on 4 plots and state if the variety differences are significant.

Plot of Land	Per acre production data		
	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

Use ANOVA, given at 5% level $F_{2,9} = 4.26$.

L3

10

OR

- Q.10 a Set up an analysis of variance table for the following two-way design results: per acre production data of wheat in metric tons;

Varieties of fertilizers	Varieties of seeds		
	A	B	C
W	6	5	5
X	7	5	4
Y	3	3	3
Z	8	7	4

Also state whether variety differences are significant at 5% level. Given that $F_{2,6} = 5.14$ and $F_{3,6} = 4.76$.

L3

10

- b Analyze the variance in the following table Latin square of yields in kgs of Paddy where A, B, C, D denotes the different methods of cultivation.

D-122	A-121	C-123	B-122
B-124	C-123	A-122	D-125
A-120	B-119	D-120	C-121
C-122	D-123	B-121	A-122

Examine whether the different methods of cultivation have given significantly different is given that $F_3 = 4.76$.

L3

10

Model Question Paper-II (CBCS Scheme)

USN

Third Semester B.E Degree Examination

MATHEMATICS FOR CS ENGINEERING STREAM (BCS301)

TIME: 03 Hours

Max.Marks:100

Note: (i) Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
(ii) Statistical tables and Mathematics Formula handbooks are allowed.

Module -1				M	L	C												
Q.01	a	A random variable X has the following probability function for various values of x			6	L2	CO1											
		<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>P(x)</td><td>0</td><td>K</td><td>2k</td><td>2k</td><td>3k</td><td>k²</td><td>2k²</td><td>7k² + k</td></tr></table>	x	0				1	2	3	4	5	6	7	P(x)	0	K	2k
x	0	1	2	3	4	5	6	7										
P(x)	0	K	2k	2k	3k	k ²	2k ²	7k ² + k										
		(i) Find the value of k (ii) Evaluate $P[X < 6]$, $P[0 < X < 5]$, and $P[X \geq 6]$																
	b	Find the mean and variance of Poisson distribution.			7	L2	CO1											
	c	In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for? (i) 10 minutes or more (ii) less than 10 minutes (iii) between 10 and 12 minutes.			7	L3	CO1											
OR																		
Q.02	a	Determine the value k, so that the function $f(x) = k(x^2 + 4)$, for $x = 0, 1, 2, 3$ can serve as a probability distribution of the discrete random variable X: Also, find (i) $P[0 < X \leq 2]$ & (ii) $P[X \geq 1]$			6	L2	CO1											
	b	Out of 800 families with 5 children each, how many would you expect to have (i) 3 boys (ii) At least one boy (iii) At most two boys, assuming equal probabilities for boys and girls.			7	L3	CO1											
	c	In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and Variance of the distribution.			7	L3	CO1											
Module-2																		
Q.03	a	If the joint probability distribution of X and Y is given by $f(x, y) = \frac{x + y}{30}, \text{ for } x = 0, 1, 2, 3; y = 0, 1, 2$ Find (i) $P[X \leq 2, Y = 1]$, (ii) $P[X > 2, Y \leq 1]$ (iii) $P[X > Y]$			6	L2	CO2											

	b	Find the unique fixed probability vector for the regular stochastic matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$	7	L2	CO2
	c	A gambler's luck follows a pattern. If he wins a game the Probability of winning the next game is 0.6. However, if he loses a game, the probability of losing the next game is 0.7. There is an even chance of the gambler winning the first game. (i) What is the probability of he winning the second game? (ii) What is the probability of he winning the third game? (iii) In the long run, how often he will win?	7	L3	CO2
OR					
Q.04	a	Determine the value of k so that the function $f(x, y) = k x - y $, for $x = -2, 0, 2; y = -2, 3$ represents joint probability distribution of the random variables X and Y. Also determine $Cov(X, Y)$.	6	L2	CO2
	b	Show that the matrix $\begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$ is a regular stochastic matrix	7	L2	CO2
	c	Three boys A, B and C are throwing a ball to each other. A is just as likely to throw the ball to B as to C. B always throws the ball to A, and C is just as likely to throw the ball to A as to B. Find the probability that C has the ball after three throws if now A has the ball.	7	L3	CO2
Module-3					
Q. 05	a	Explain the following terms (i) Standard error (ii) Statistical hypothesis (iii) Critical region of a statistical test (iv) Test of significance	6	L1	CO3
	b	In 324 throws of a six faced die, an odd number turned up 181 times. Is it reasonable to think that the die is unbiased one at 5% level of significance?	7	L3	CO3
	c	In an examination given to students at a large number of different schools the mean grade was 74.5 and S.D grade was 8. At one particular school where 200 students took the examination the mean grade was 75.9. Discuss the significance of this result at both 5% and 1% level of significance.	7	L3	CO3
OR					
Q. 06	a	Define (i) Alternative hypothesis (ii) A statistic (iii) Level of significance and (iv) Two-tailed test	6	L1	CO3

	b	A coin is tossed 1000 times and head turns up 540 times. Decide on the hypothesis that the coin is unbiased at 1% level of significance.	7	L3	CO3																					
	c	One type of air craft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significance difference in the two types of air craft's so far as engine defects are concerned? Test at 5% significance level.	7	L3	CO3																					
Module-4																										
Q. 07	a	An unknown distribution has a mean of 90 and a standard deviation of 15. Samples of size $n = 25$ are drawn randomly from the population. Find the probability that the sample mean is between 85 and 92.	6	L2	CO4																					
	b	The heights of a random sample of 50 college students showed a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. Construct a 99% confidence interval for the mean height of all college students.	7	L2	CO4																					
	c	A die was thrown 60 times and the following frequency distribution was observed: <table border="1"><tr><td>Faces</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>frequency</td><td>15</td><td>6</td><td>4</td><td>7</td><td>11</td><td>17</td></tr></table> Test whether the die is unbiased at 5% significance level.	Faces	1	2	3	4	5	6	frequency	15	6	4	7	11	17	7	L3	CO4							
Faces	1	2	3	4	5	6																				
frequency	15	6	4	7	11	17																				
OR																										
Q. 08	a	In a recent study reported on the Flurry Blog, the mean age of tablet users is 34 years. Suppose the standard deviation is 15 years. Take a sample of size $n = 100$. Using central limit theorem, find the probability that the sample mean age is more than 30 years.	6	L2	CO4																					
	b	Suppose that 10, 12, 16, 19 is a sample taken from a normal population with variance 6.25. Find a 95 percent confidence interval for the population mean.	7	L2	CO4																					
	c	A random sample of 10 boys had the following I.Q.: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100 (at 5% level of significance)?	7	L3	CO4																					
Module-5																										
Q. 09	a	Three types of fertilizers are used on three groups of plants for 5 weeks. We want to check if there is a difference in the mean growth of each group. Using the data given below apply a one-way ANOVA test at 0.05 significant level <table border="1"><tr><td>Fertilizer-1</td><td>6</td><td>8</td><td>4</td><td>5</td><td>3</td><td>4</td></tr><tr><td>Fertilizer-2</td><td>8</td><td>12</td><td>9</td><td>11</td><td>6</td><td>8</td></tr><tr><td>Fertilizer-3</td><td>13</td><td>9</td><td>11</td><td>8</td><td>7</td><td>12</td></tr></table>	Fertilizer-1	6	8	4	5	3	4	Fertilizer-2	8	12	9	11	6	8	Fertilizer-3	13	9	11	8	7	12	10	L3	CO5
Fertilizer-1	6	8	4	5	3	4																				
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	b	A coin is tossed 1000 times and head turns up 540 times. Decide on the hypothesis that the coin is unbiased at 1% level of significance.	7	L3	CO3																					
	c	One type of air craft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significance difference in the two types of air craft's so far as engine defects are concerned? Test at 5% significance level.	7	L3	CO3																					
Module-4																										
Q. 07	a	An unknown distribution has a mean of 90 and a standard deviation of 15. Samples of size $n = 25$ are drawn randomly from the population. Find the probability that the sample mean is between 85 and 92.	6	L2	CO4																					
	b	The heights of a random sample of 50 college students showed a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. Construct a 99% confidence interval for the mean height of all college students.	7	L2	CO4																					
	c	A die was thrown 60 times and the following frequency distribution was observed: <table border="1"><tr><td>Faces</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>frequency</td><td>15</td><td>6</td><td>4</td><td>7</td><td>11</td><td>17</td></tr></table> Test whether the die is unbiased at 5% significance level.	Faces	1	2	3	4	5	6	frequency	15	6	4	7	11	17	7	L3	CO4							
Faces	1	2	3	4	5	6																				
frequency	15	6	4	7	11	17																				
OR																										
Q. 08	a	In a recent study reported on the Flurry Blog, the mean age of tablet users is 34 years. Suppose the standard deviation is 15 years. Take a sample of size $n = 100$. Using central limit theorem, find the probability that the sample mean age is more than 30 years.	6	L2	CO4																					
	b	Suppose that 10, 12, 16, 19 is a sample taken from a normal population with variance 6.25. Find a 95 percent confidence interval for the population mean.	7	L2	CO4																					
	c	A random sample of 10 boys had the following I.Q.: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100 (at 5% level of significance)?	7	L3	CO4																					
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	b	Present your conclusions after doing analysis of variance to the following results of the Latin-square design experiment conducted in respect of five fertilizers which were used on plots of different fertility	10	L3	CO5																																																		
		<table><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>16</td><td>10</td><td>11</td><td>9</td><td>9</td></tr><tr><td>E</td><td>C</td><td>A</td><td>B</td><td>D</td></tr><tr><td>10</td><td>9</td><td>14</td><td>12</td><td>11</td></tr><tr><td>B</td><td>D</td><td>E</td><td>C</td><td>A</td></tr><tr><td>15</td><td>8</td><td>8</td><td>10</td><td>18</td></tr><tr><td>D</td><td>E</td><td>B</td><td>A</td><td>C</td></tr><tr><td>12</td><td>6</td><td>13</td><td>13</td><td>12</td></tr><tr><td>C</td><td>A</td><td>D</td><td>E</td><td>B</td></tr><tr><td>13</td><td>11</td><td>10</td><td>7</td><td>14</td></tr></table>				A	B	C	D	E	16	10	11	9	9	E	C	A	B	D	10	9	14	12	11	B	D	E	C	A	15	8	8	10	18	D	E	B	A	C	12	6	13	13	12	C	A	D	E	B	13	11	10	7	14
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Q. 10	a	A trial was run to check the effects of different diets. Positive numbers indicate weight loss and negative numbers indicate weight gain. Check if there is an average difference in the weight of people following different diets using an ANOVA Table.	10	L3	CO5																																																		
		<table><tr><td>Low Fat</td><td>Low Calorie</td><td>Low protein</td><td>Low carbohydrate</td></tr><tr><td>8</td><td>2</td><td>3</td><td>2</td></tr><tr><td>9</td><td>4</td><td>5</td><td>2</td></tr><tr><td>6</td><td>3</td><td>4</td><td>-1</td></tr><tr><td>7</td><td>5</td><td>2</td><td>0</td></tr><tr><td>3</td><td>1</td><td>3</td><td>3</td></tr></table>				Low Fat	Low Calorie	Low protein	Low carbohydrate	8	2	3	2	9	4	5	2	6	3	4	-1	7	5	2	0	3	1	3	3																										
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	b	The following data show the number of worms quarantined from the GI areas of four groups of muskrats in a carbon tetrachloride anthelmintic study. Conduct a two-way ANOVA test.	10	L3	CO5																																																		
		<table><tr><td>I</td><td>II</td><td>III</td><td>IV</td></tr><tr><td>33</td><td>41</td><td>12</td><td>38</td></tr><tr><td>32</td><td>38</td><td>35</td><td>43</td></tr><tr><td>26</td><td>40</td><td>46</td><td>25</td></tr><tr><td>14</td><td>23</td><td>22</td><td>13</td></tr><tr><td>30</td><td>21</td><td>11</td><td>26</td></tr></table>				I	II	III	IV	33	41	12	38	32	38	35	43	26	40	46	25	14	23	22	13	30	21	11	26																										
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